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Case Report

Evaluating the efficiency of nanofiltration and reverse osmosis membranes for the removal of micro-pollutants using a machine learning approach

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ABSTRACT

Water removal research is a critical study area across diverse industries and scientific domains. This investigation evaluates three prominent machine learning models including Artificial Neural Networks, Random Forest, and XGBoost for their predictive efficacy concerning water removal efficiency in reverse osmosis (RO) and nanofiltration processes. The study aims to assess the influence of molecular weight (MW) and compound size on removal rates within RO and nanofiltration contexts. After training with an extensive dataset, the Artificial Neural Network (ANN) model was created using a four-layer architecture with 128, 64, 32, and 32 nodes over 2500 epochs. Notably, the ANN model exhibited superior performance, achieving an R^2 score of 0.9795, indicating its precision in elucidating the variability observed within RO and nanofiltration processes. The Random Forest model (R^2: 0.9658) and XGBoost model (R^2: 0.9447) also demonstrated substantial predictive capabilities. These results offer valuable insights for optimizing water removal techniques within these processes.

1. Introduction

Membrane-based water treatment technologies have emerged as a promising solution for removing micro-pollutants from water sources [1,2]. These contaminants, which include organic micropollutants, herbicides, and salts, pose significant environmental and health risks [3]. Researchers have recently focused on optimizing membrane processes to enhance their efficiency and effectiveness in micropollutant removal. Several related studies have contributed to the understanding and advancement of membrane-based water treatment technologies for micropollutant removal [4]. One study investigated the efficiency of ultrafiltration and nanofiltration methods in removing phenyl-urea herbicides from ultrapure water, highlighting the superior removal rates of nanofiltration due to its smaller pore size [5]. Another study explored incorporating hydrophilic carbon dots into polyamide thin film nanocomposite reverse osmosis membranes, improving hydrophilicity, anti-fouling properties, and increased water flow [6]. Likewise, the integration of graphene oxide into thin film composite membranes demonstrated its effectiveness in salt removal from seawater [7]. Furthermore, a study emphasized the potential of nanofiltration and reverse osmosis in removing organic micropollutants from water and wastewater, highlighting various parameters that influence process efficiency [8]. Another investigation used annealed polyelectrolyte multilayer-based nanofiltration membranes for municipal wastewater treatment, demonstrating high micropollutant rejection rates [9]. Lastly, the importance of developing acid-resistant nanofiltration membranes for fluoride removal applications was underscored [10]. The review by Vasseghian et al. [11] highlights the importance of these membrane processes, showcasing various parameters influencing their efficiency. Additionally, Tran et al. [12] focus on annealed polyelectrolyte multilayer-based nanofiltration membranes for municipal wastewater treatment, demonstrating promising micropollutant rejection rates. Furthermore, Vasseghian et al. underscores the need for acid-resistant nanofiltration membranes in fluoride removal applications [13]. These studies collectively contribute to understanding membrane technologies' role in addressing micropollutant challenges, providing insights into parameters and membrane designs, and enhancing removal efficiency. This research paper explores the potential of using machine learning (ML) algorithms in designing and optimizing membrane-based water treatment systems for micropollutant removal. By analyzing the existing research and incorporating machine learning algorithms (Table 1), our study aims to contribute to the optimization of membrane processes for micropollutant removal. In this study, we leveraged Artificial Neural Networks (ANN), Random Forest, and

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Table 1
Detailed data acquired after the detection of outliers.

	Compound Molecular Weight (g/mol)	Pressure (kPa)	Measurement time (min)	Initial concentration (mg/L)	Name of compound	Compound size (nm)	Compound Kow
Count..	1802.00	1802.00	1802.00	1802.00	–	1802.00	1802.00
Mean	246.53	850.96	117.74	5.88	polyethylene (PE)	0.52	1.80
Std..	84.19	487.92	189.82	6.24	polysulfone (PSf)	0.8	1.47
Min..	46.06	276.00	10.00	0.01	Poly (methyl methacrylate) (PMMA)	0.27	−2.30
25 %	200.62	500.00	10.00	0.10	polyvinylidene fluoride (PVDF)	0.48	1.15
50 %	236.31	1000.00	10.00	1.56	polyacrylonitrile (PAN)	0.54	1.88
75 %	296.15	1000.00	120.00	10.00	Polyimide (PI)	0.58	2.76
Max	446.48	3500.00	720.00	20.00	Polyallylamine (PAA)	0.72	5.70

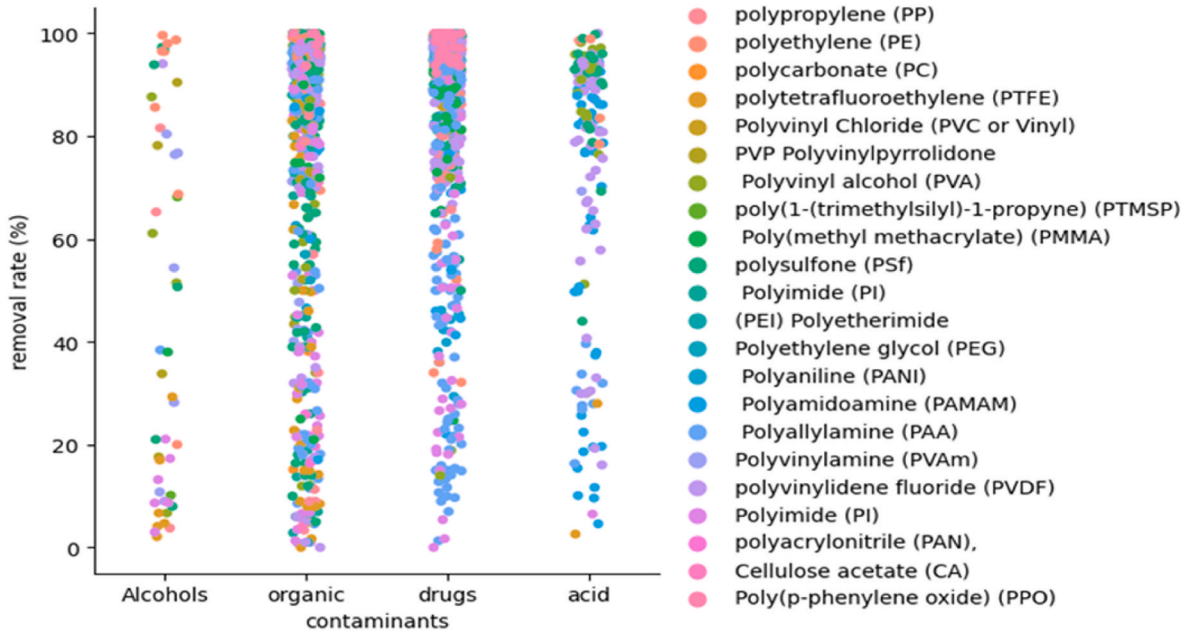


Fig. 1. Dispersion of membrane types and their removal rates for different pollutants.

XGBoost as machine learning models for evaluating the efficiency of nanofiltration and reverse osmosis membranes. These models were selected for their distinct advantages in handling complex, nonlinear relationships within the dataset. We collected data on micropollutant removal from 45 studies, specifically on lab-based polymer membranes, to ensure a consistent data set [14]. These models allowed us to consider various influential factors simultaneously, including membrane material properties (e.g., pore size) and micropollutant characteristics (e.g., molecular weight, hydrophobicity), to develop more accurate predictions of the efficacy of nanofiltration (NF) and reverse osmosis (RO) membranes in micropollutant removal [15]. With its ability to capture intricate patterns and relationships, ANN was chosen to uncover nuanced interactions between various parameters influencing micro-pollutant removal. Random Forest, known for its robustness against overfitting and interpretability, was employed to provide insights into the importance of individual features. XGBoost, a powerful ensemble learning technique, was utilized to further enhance predictive accuracy by leveraging the strengths of gradient boosting. By employing this combination of machine learning models, we aimed to comprehensively assess the efficiency of nanofiltration and reverse osmosis membranes for micro-pollutant removal. The developed prediction models can guide the selection of appropriate membrane materials and operating conditions to maximize removal efficiency, thereby mitigating the environmental and health impacts of micropollutants in water sources.

2. Materials and method

2.1. Data collection

To collect the necessary data for our study on micropollutant removal by nanofiltration and reverse osmosis membranes, we thoroughly searched various keywords related to these membranes and the names of common polymer materials used in such membranes. The search conducted resulted in the identification of 45 articles that contained pertinent data for our investigation. From these articles, we were able to identify six critical characteristics that we considered would be most significant to our research. These variables were molecular weight, the octanol-water partition coefficient, Pressure, initial concentration, measurement time, and compound size. The dataset used six variables and consisted of 1905 individual data points. The dataset included 23 different polymeric membranes and four primary categories of contaminants, including organic, pharmaceutical, alcohol, and acid (Fig. 1). Our objective was to comprehensively comprehend the factors that affect the elimination of micropollutants by nanofiltration and reverse osmosis membranes by incorporating a diverse range of compounds and operating conditions into our dataset. The present study elucidates the pivotal significance of the six variables in eliminating micropollutants and can provide valuable insights for developing and enhancing membrane filtration techniques.

The molecular weight (MW) is defined as the numerical value



Fig. 2. Pearson correlation matrix between any two properties and between removal rate and each variable.

representing the mass of a molecule, which is commonly expressed in atomic mass units (AMU). The present investigation centers on the MW of diverse compounds and their impact on their elimination rates in nanofiltration and reverse osmosis membranes. The octanol-water partition coefficient (Kow) measures a compound's hydrophobicity, or proclivity to partition into nonpolar liquids. It is computed by dividing the compound's concentration in octanol by its concentration in water. To evaluate how hydrophobicity affects chemical removal rates in membranes, we incorporated Kow values into our investigation [16]. Pressure is critical in membrane separation processes, such as nanofiltration and reverse osmosis. It directly affects removal rates by influencing the driving force for solute transport through membrane pores. Understanding the impact of pressure helps optimize membrane performance and enhance removal efficiencies. The initial concentration of a compound is the amount of the compound present in a solution before it is introduced to a membrane. This variable was included in our study to investigate how varying initial concentrations of different compounds may affect their removal rates in membranes [17]. The measurement time refers to the duration of time over which a compound is exposed to a membrane before the concentration of the compound is measured. We included this variable to see how the removal rate of compounds changes over time and to identify the optimal exposure time for membrane filtration [18]. Finally, compound size refers to the size of the compound being filtered by the membrane. We included this variable to investigate how the size of different compounds affects their removal rates in membranes with different MWCOs. By examining these six variables, we aimed to gain a deeper understanding of the factors that affect micropollutant removal by nanofiltration and reverse osmosis membranes [19].

2.2. Data preprocessing

Data preprocessing is a critical step in any data analysis, and this study was no exception. Before doing any analysis on the dataset, many preprocessing processes are done to clean and prepare it. First, the dataset was verified for missing values and any data points with incomplete information. Then, reviewed the data for outliers or errors and fixed or eliminated them as appropriate. To identify outliers, the Z-score method was utilized, which involves calculating the Z-score for each data point to measure its deviation from the mean.

$$Z = (X - \mu) / \sigma \quad (1)$$

Where Z is the Z-score of the data point, X is the value of the data point. μ is the mean of the dataset. σ is the standard deviation of the dataset. A threshold value, often 2 or 3, is typically set to identify outliers using the Z-score method. If the Z-score of a data point exceeds this threshold, it is considered an outlier. The data was then normalized to verify that all variables had a mean of zero and a standard deviation of one. This step is critical because it helps to reduce the impact of various measurement scales on the analysis [20].

$$X_{normalized} = (X - X_{mean}) / X_{std} \quad (2)$$

A quantitative analysis of features throughout the data preparation phase was performed to determine the most relevant factors in predicting micropollutant removal rates. Pearson correlation matrix analysis was one of the statistical approaches used. Pearson correlation analysis assesses the strength and direction of a variable's linear connection. The degree of relationship was measured by generating correlation coefficients between each variable and the micropollutant clearance rates. Variables exhibiting higher correlation coefficients demonstrate a more robust association with removal rates, indicating their potential significance in predictive models. Factors with substantial associations with micropollutant clearance rates were selected based on Pearson correlation matrix analysis. These variables, identified as the most important characteristics in the dataset due to their significant

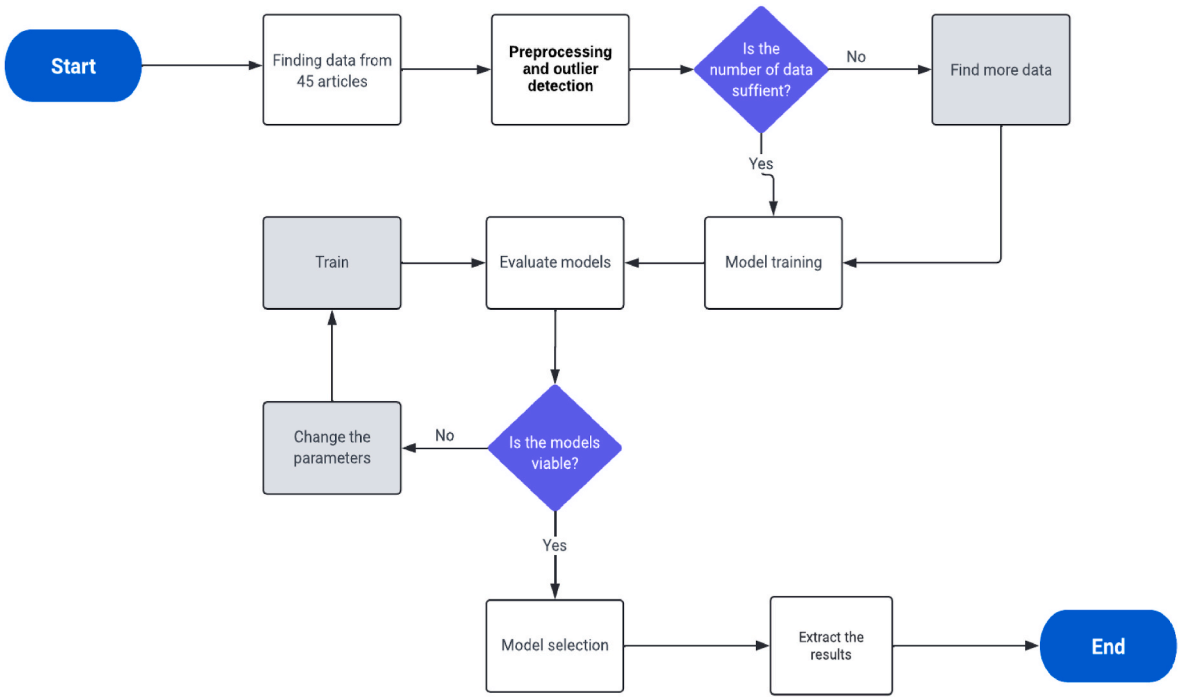


Fig. 3. The present machine learning-based modeling procedure and algorithm diagram.

statistical association with the target variable (Fig. 2), were then subjected to regression analysis. This analysis assessed the individual contributions of these chosen factors in predicting micropollutant removal rates. The influence of each variable on the removal rates was quantified through fitting regression models, and their importance in the prediction models was evaluated. This investigation facilitated the development of prediction models by offering insights into the relative importance of each attribute. In summary, quantitative feature analysis and Pearson correlation matrix analysis identified relevant factors and their correlations with micropollutant removal rates. This approach allowed for a focused examination of the essential elements impacting membrane filtration's efficacy in removing various micropollutants. The dataset was divided into training and validation sets to evaluate model performance. Ninety percent of the data in the training set was allocated for model training and parameter tuning. At the same time, the remaining 10 % was reserved as a validation set to assess the models' predicted accuracy on unseen data. By evaluating their performance on samples not included in the training phase, it was possible to ascertain the models' generalization to new data. This process helped identify whether the models were overfitting or capturing relevant trends in the data. The performance of the models on unknown data can be evaluated using a separate validation set, allowing for necessary adjustments to enhance their generalization capabilities. This approach ensured a consistent evaluation of the model's prediction accuracy and ability to handle new data instances. In summary, these preprocessing techniques were instrumental in ensuring the research was conducted on high-quality data and that the models were reliable and accurate [21].

2.3. Model selection

In the model selection phase, the objective is to choose the most appropriate machine learning algorithm that yields the best performance for predicting micropollutant removal rates based on our dataset and problem requirements. This involves evaluating and comparing the performance of various candidate models. As part of our model selection process, we explored three machine learning algorithms: Random Forest Regressor, XGBoost, and Neural Network. Each algorithm offers unique capabilities for predicting micropollutant removal rates, carefully evaluating their performance, and fine-tuning their hyperparameters to ensure optimal results (Fig. 3).

2.3.1. Random Forest Regressor

The Random Forest Regressor utilizes an ensemble of decision trees to predict outcomes by effectively managing complex relationships and capturing non-linear patterns within the data [22]. This ensemble approach reduces overfitting and enhances the model's ability to generalize. Key hyperparameters for this regressor include the number of trees, maximum tree depth, and the minimum samples needed to split an internal node during tree construction [23].

2.3.2. XGBoost (extreme gradient boosting)

XGBoost, an implementation of gradient boosting, sequentially constructs an ensemble of weak prediction models. Renowned for its high performance and computational efficiency, XGBoost excels in handling complex feature interactions and employs regularization techniques. Essential hyperparameters for XGBoost encompass the number of boosting rounds, learning rate, maximum tree depth, and regularization parameters such as gamma, lambda, and alpha [24].

2.3.3. Neural network (multilayer perceptron)

The Multilayer Perceptron (MLP) is a pivotal artificial neural network architecture featuring interconnected layers of nodes or neurons [25]. Each neuron processes the weighted sum of its inputs, applies an activation function, and transfers the result to the next layer [26]. The MLP is extensively used for solving complex problems in machine learning. Calculation of the weighted sum (net) involves multiplying inputs (x) by corresponding weights (w), summing them, and adding a bias term (b):

$$\text{net}_i = \sum(w_{ij} \times x_j) + b_i \quad (3)$$

The net value then undergoes an activation function (f), introducing nonlinearity and transforming it into the neuron's output:

$$\text{output}_i = f(\text{net}_i) \quad (4)$$

Activation functions like sigmoid, tanh, and ReLU introduce non-

Table 2
Calculation of adsorption parameters by models by fitting the experimental data for tapes of membrane (MB).

Type of MB	Contaminant	Compound Mw (g/mol)	Compound Kow	Pressure (kpa)	Measurement Time (min)	Initial concentration (mg/L	Compound size (nm)	Removal rate (%)	ANN	XGBoost	Random Forest
PVDF	organic	162.5	0.711	550	300	235	0.87	99	99.3	94.3	97.5
PsF	Acid	113	0.31	400	250	40	0.66	87	86.9	89.4	86.4
PAA	Pharmaceutical	60	0.255	860	55	1670	0.11	67	67.1	66	67.1

linearity. Stacking layers of neurons and applying these computations empowers MLPs to learn intricate patterns within data. During training, weights and biases are adjusted to optimize performance [27]. Key hyperparameters for Neural Networks involve the architecture's hidden layer specifics, activation functions, learning rate, batch size, and the number of training epochs [24]. Hyperparameter tuning was accomplished through grid or random search techniques, exploring various hyperparameter combinations. This enabled the identification of the optimal configurations for each algorithm, resulting in superior performance on our dataset [28]. To determine the best-performing model, the performance of these algorithms was compared using appropriate evaluation metrics such as mean squared error (MSE), mean absolute error (MAE), or R-squared (R^2) score. Mean Absolute Error (MAE): The Mean Absolute Error measures the average magnitude of errors between predicted and actual values [22]. It provides a measure of the model's accuracy (Table 2).

Formula:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (5)$$

where n is the total number of data points, y represents the actual values, ŷ represents the predicted values. Mean squared error (MSE): The mean squared error measures the average squared difference between predicted and actual values. It emphasizes larger errors than MAE and is commonly used in regression analysis.

Formula:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \quad (6)$$

Root Mean Squared Error (RMSE): The root mean squared error is the square root of the mean squared error. It provides a measure of the standard deviation of the errors and is useful for comparing different models [29].

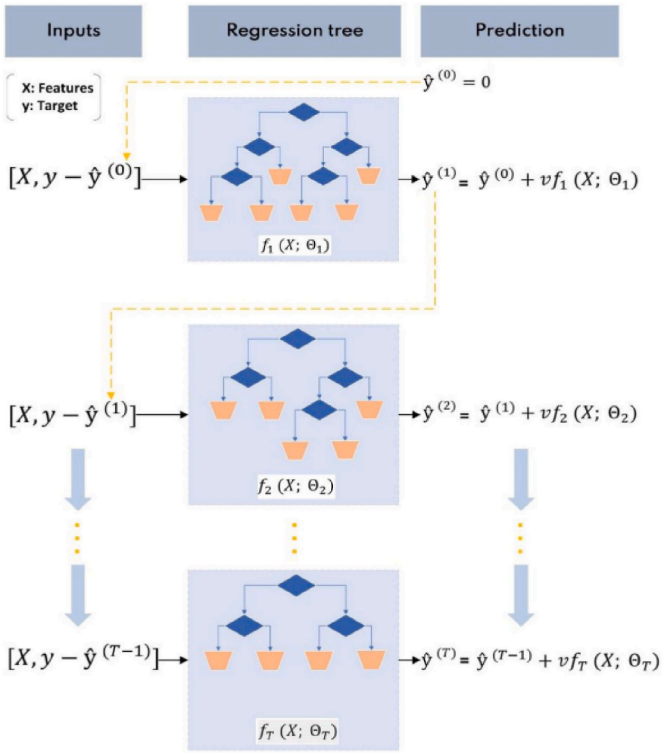


Fig. 5. Schematic diagram of gradient boosting procedure [30].

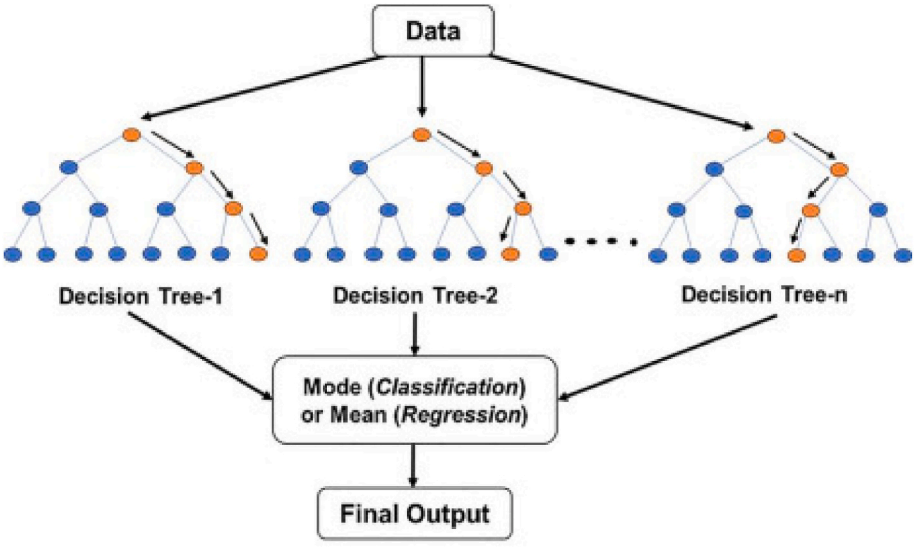


Fig. 4. Schematic diagram of random forest procedure.

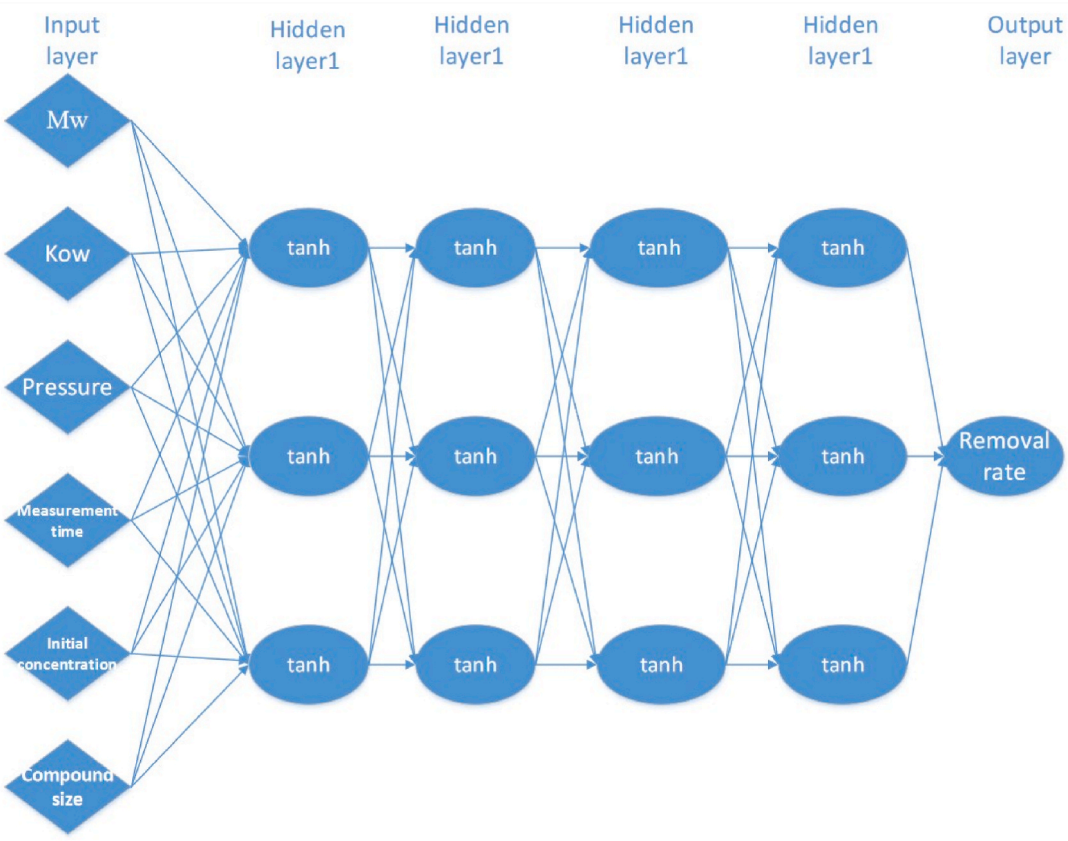


Fig. 6. Schematic diagram of multi-layer perceptron (MLP) network.

Formula:

$$\text{the RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad 7$$

R-squared (R^2) Score: The R-squared score represents the proportion of the variance in the dependent variable that is predictable from the independent variables. It indicates the model's goodness of fit.

Formula:

$$R^2 = \frac{\sum_{i=1}^{n}(Y_{predicted} - Y_{actual})^2}{\sum_{i=1}^{n}(Y_{predicted} - Y_{mean})^2} \quad 8$$

Where:

- o SSR (Sum of Squares Residuals) measures the sum of squared differences between predicted and actual values.
- o SST (Total Sum of Squares) measures the sum of squared differences between the actual values and the mean of the dependent variable [24].

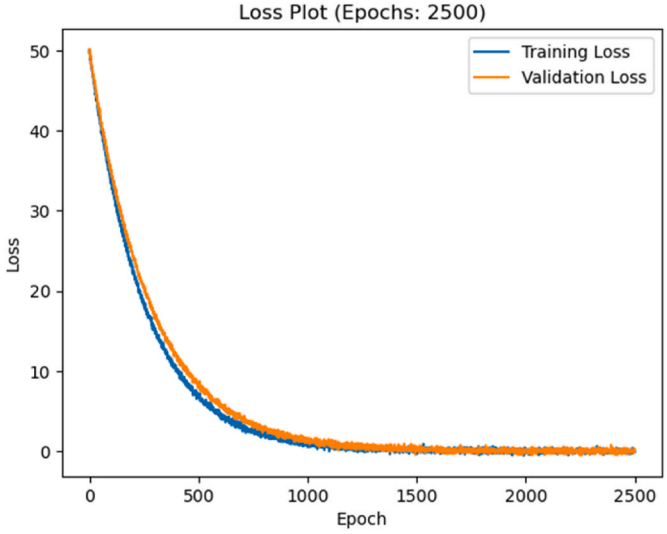


Fig. 7. The learning curve of the multi-layer perceptron model.

3. Result and discussion

In this study, grid search was employed to tune the hyperparameters of the three models: Random Forest Regressor (Fig. 4), XGBoost (Fig. 5), and Neural Network-MLP (Fig. 6). Grid search is a technique that exhaustively searches the specified parameter grid to identify a given model's optimal combination of hyperparameters.

Specific hyperparameters for each model were employed during this study. For the Random Forest Regressor, 700 trees with a maximum depth of 10 were utilized, requiring a minimum of 2 samples to split an internal node and a minimum of 1 sample to be at a leaf node. For the XGBoost model, 500 boosting rounds with a maximum tree depth of 6 were conducted, utilizing a learning rate of 0.1 and a training instance subsample ratio of 0.8. The Neural Network-MLP was configured with four hidden layers containing 128, 64, 32, and 32 nodes, respectively, trained for 2500 epochs with a learning rate of 0.0001 and a batch size of 32. These hyperparameters were selected based on the grid search evaluation, which systematically explored different combinations of hyperparameter values to identify the optimal configuration for each model (Fig. 7). Table 2 presents the model evaluation results using the MSE and R^2 metrics. The ANN demonstrated the best fit to the experimental data among the three models, with an MSE of 0.024 and a

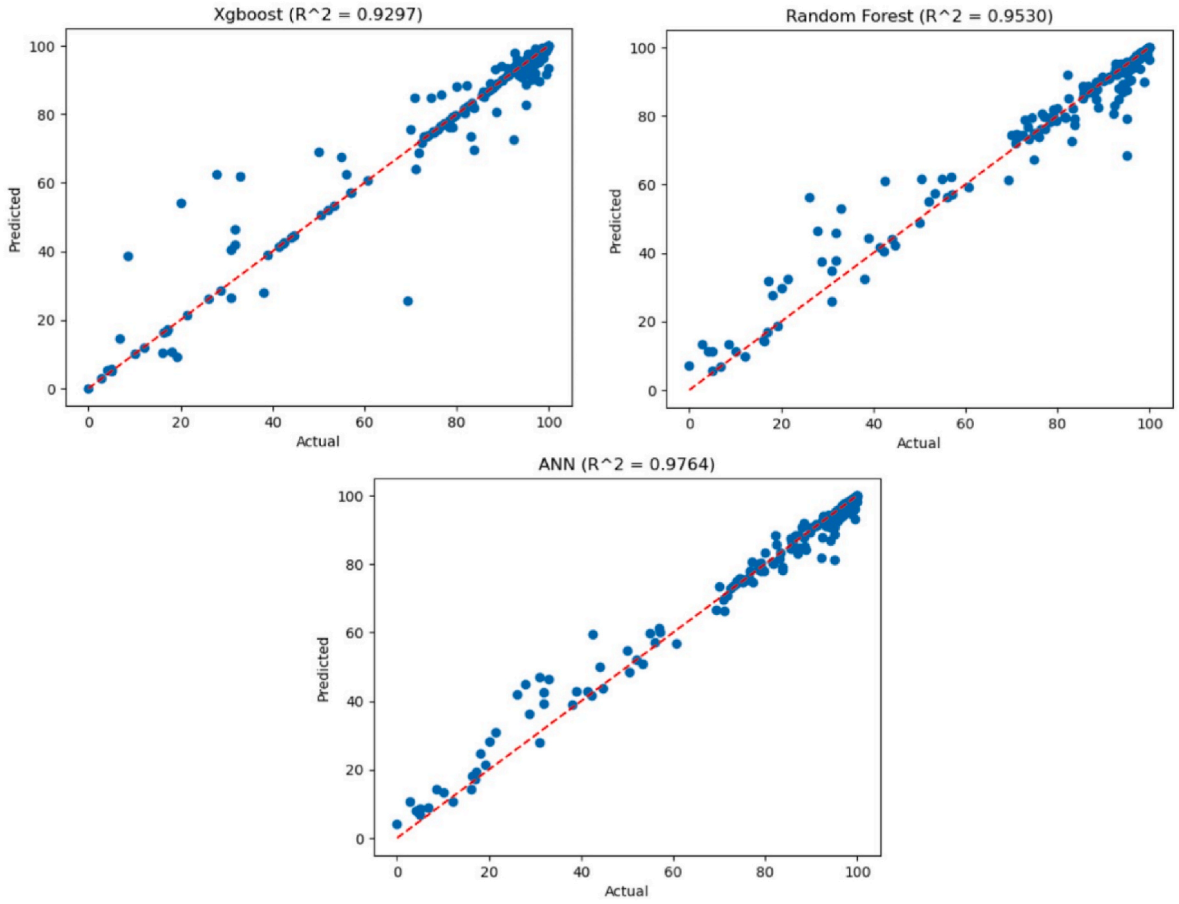


Fig. 8. Removal rate experimental versus predicted data using Artificial neural network, XGboost, and Random Forest models.

Table 3
Analytical criteria for model comparison.

Model	Mean Squared Error (MSE)	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R^2
ANN	0.024	0.139	0.155	0.9795
Random Forest	0.103	0.322	0.321	0.9658
XGBoost	0.135	0.275	0.368	0.9447

maximum R^2 value of 0.9795. The Random Forest model achieved an MSE of 0.103 and a maximum R^2 value of 0.9658, while the XGBoost model had an MSE of 0.135 and a maximum R^2 value of 0.9447. Based on the evaluation results shown in Table 2, it was determined that the ANN model provided the most accurate predictions of the containment removal rates, as indicated by its lower MSE and higher R^2 values than the other model (Fig. 8). To further validate the performance of the optimal ANN model, some experimental values of containment removal rates from the literature were randomly selected and compared with the model's predicted values. The results of this comparison are presented in the subsequent section, providing additional evidence of the model's predictive capabilities (Table 3). These variations in performance can be attributed to several key factors. As a deep learning model, the ANN can capture intricate patterns within the data due to its inherent complexity. This likely contributed to its superior performance, resulting in lower MSE and higher R^2 values than the Random Forest and XGBoost models. Additionally, each machine learning model interprets feature representations differently. The ANN's ability to learn hierarchical feature representations may have affected its effectiveness. Furthermore, the performance of machine learning models heavily relies on hyperparameter tuning. While all models were fine-tuned, the ANN may have benefited from an optimal configuration that led to its superior performance metrics. The size and quality of the dataset can also influence model performance. The ANN may have had ample data to learn complex patterns, potentially outperforming the Random Forest and XGBoost models. Lastly, considerations of overfitting are crucial. The ANN may have been less susceptible to overfitting due to its architecture and regularization techniques, resulting in more generalized predictions. In this study, Mean Squared Error (MSE) and R-squared (R^2) metrics were utilized to evaluate the performance of machine learning models in predicting removal rates within nanofiltration and reverse osmosis processes. MSE, a common regression metric, calculates the average squared difference between predicted and actual values, offering insights into prediction accuracy, with lower values indicating greater precision but sensitivity to outliers. R^2 measures the goodness of fit, representing the proportion of variance in removal rates explained by the variables, though it does not address bias. These metrics were chosen for their relevance to regression tasks and their capacity to provide an understanding of model accuracy and explanatory capability. Minimizing MSE aimed to enhance prediction accuracy, while high R^2 values indicated better alignment of models with the data. The rationale for selecting these metrics and their respective strengths and limitations has been detailed in the Methods section to clarify the decision-making process and align with the editor's feedback for improved transparency in the model evaluation process. In conclusion, the ANN model's superior performance, evidenced by its lower MSE and higher R^2 values, can be attributed to its ability to handle complex patterns, effective feature representation, optimal hyperparameter tuning, dataset size, and potential mitigation of overfitting.

To visualize the relationships between the variables and the removal

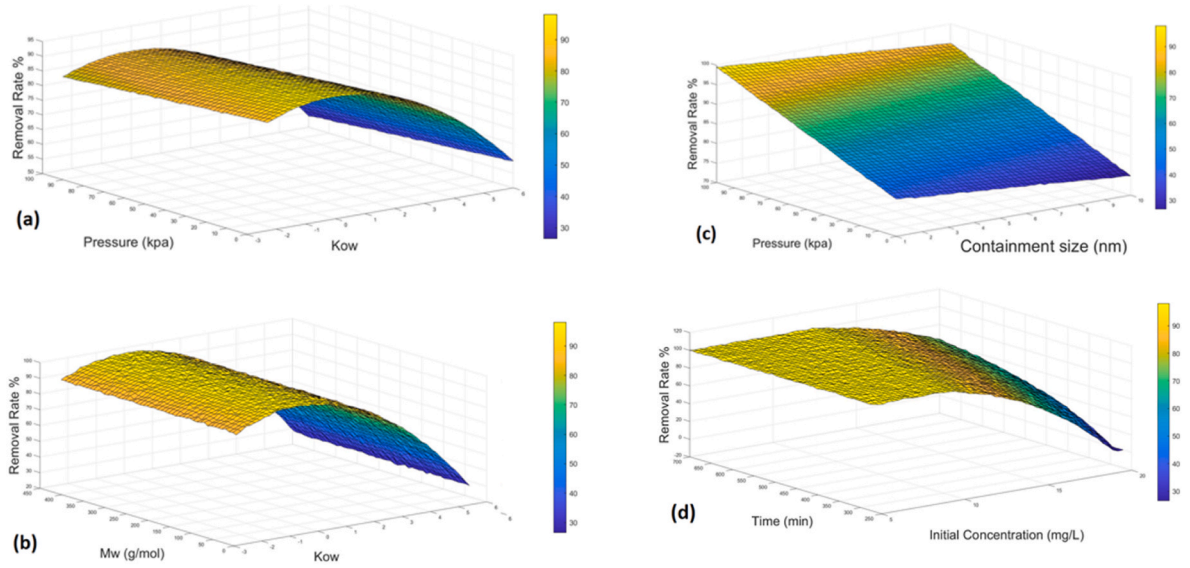


Fig. 9. a) Influence of Kow and pressure on the removal rate, b) the interplay between Kow and MW on the removal rate, c) the relationship between pore size and pressure on the removal rate, d) impact of initial concentration and time on the removal rate generated by ANN model.

rates more effectively, three-dimensional diagrams were utilized for the ANN model. These diagrams explore the combined effects of multiple variables on the removal rates in a comprehensive manner. Firstly, the influence of Kow (octanol-water partition coefficient) and pressure on the removal rate (Fig. 9a) were investigated. Results demonstrated a positive correlation between Kow and the removal rate, indicating that compounds with higher Kow values exhibit higher removal rates. Additionally, the observation that increased pressure led to higher removal rates, suggesting that pressure plays a crucial role in enhancing removal efficiency. These findings emphasize the importance of considering both Kow and pressure when designing water treatment systems to achieve optimal removal of contaminants. Next, the impact of initial concentration and time on the removal rate was examined (Fig. 9d). This analysis revealed that as the initial concentration of the compound increased, the removal rate decreased. This inverse relationship suggests that higher initial concentrations may require additional treatment measures to achieve desired removal rates. Furthermore, we found that the removal rate generally increased with longer treatment times, indicating the importance of sufficient contact time for effective contaminant removal.

Moreover, the interplay between Kow and molecular weight (MW) about the removal rate was investigated (Fig. 9b). These findings indicated that compounds with higher MW and Kow values tended to exhibit lower removal rates. This suggests that larger, more hydrophobic compounds may pose challenges for efficient removal in water treatment processes. Understanding the combined influence of MW and Kow provides insights into the size exclusion effects on removal rates and aids in selecting appropriate treatment strategies. Finally, we examined the relationship between containment size and pressure on the removal rate (Fig. 9c). Our results demonstrated that increasing the containment size in conjunction with higher pressure resulted in higher removal rates. This analysis revealed that the molecular weight of the compounds and the octanol-water partition coefficient (Kow) significantly impact the removal rate. Generally, compounds with larger molecular weights and higher Kow values exhibit lower removal rates. Additionally, the influence of pressure on the removal rate of pollutants was observed, indicating that higher pressures often result in a higher removal rate.

One possible explanation for the effect of increased pressure on removal rates could be enhanced mass transfer kinetics. As pressure rises, the solubility of contaminants in the solvent also increases, facilitating their transfer from the aqueous phase to the adsorbent material. This results in a more efficient removal process, with higher pressure creating favorable conditions for increased adsorption capacity [31]. Furthermore, the inverse relationship between initial concentration and removal rate merits thorough consideration. This intriguing observation suggests that as the initial concentration of contaminants decreases, their removal rate actually increases. Such a trend could indicate the presence of saturation effects or competition for adsorption sites at higher initial concentrations.

4. Conclusion

In conclusion, our study investigated the factors influencing the removal rates of contaminants in nanofiltration and reverse osmosis membranes, with a specific focus on compound MW, compound Kow, and Pressure. Through developing and evaluating predictive models, we found that the ANN-MLP model demonstrated superior performance in estimating the removal rates, with a low MSE and high R^2 value. Our analysis revealed that compound MW and compound Kow significantly impacted the removal rates, with larger MW and higher Kow values associated with lower removal rates. Additionally, we observed that Pressure played a crucial role in determining the removal rates, with higher pressures resulting in higher removal rates. These findings emphasize the importance of considering the compound properties and membrane characteristics, such as MW and Kow, and the applied pressure when designing and optimizing membrane systems for effective contaminant removal. To visually represent these relationships, we created three-dimensional diagrams illustrating the combined influence of compound MW, compound Kow, and pressure on the removal rates. These diagrams comprehensively explained how these variables interact and affect contaminant removal in nanofiltration and reverse osmosis processes. Overall, our study contributes to the knowledge of the factors influencing contaminant removal rates in membrane filtration and highlights the potential of predictive models in accurately estimating removal rates based on the considered variables. This information can guide researchers and engineers in making informed decisions to improve the efficiency of water treatment processes.

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Ethical approval and consent to participate

Not applicable.

Consent to publish

Not applicable.

CRediT authorship contribution statement

Parnian Masuodi: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Fatemeh Bahmanzadegan:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Alireza Hemmati:** Visualization, Validation, Supervision, Conceptualization. **Ahad Ghaemi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cscee.2024.100750.

References

[1] M.F. Idrees, Performance analysis and treatment technologies of reverse osmosis plant–a case study, Case studies in Chemical and environmental engineering 2 (2020) 100007.
[2] A.H. Birniwa, et al., Membrane technologies for heavy metals removal from water and wastewater: a mini review, Case Studies in Chemical and Environmental Engineering (2023) 100538.
[3] M.Q. Gubari, H.M. Zwain, W.H. Hassan, M. Vakili, A. Majdi, Desalination of pigment industry wastewater by reverse osmosis using OPM-K membrane, Case Studies in Chemical and Environmental Engineering 8 (2023) 100401.
[4] A. Azanaw, B. Birlie, B. Teshome, M. Jemberie, Textile effluent treatment methods and eco-friendly resolution of textile wastewater, Case Studies in Chemical and Environmental Engineering 6 (2022) 100230.
[5] F.J. Benitez, F.J. Real, J.L. Acero, C. Garcia, Kinetics of the transformation of phenyl-urea herbicides during ozonation of natural waters: rate constants and model predictions, Water Res. 41 (18) (2007) 4073–4084.
[6] Y. Li, S. Li, K. Zhang, Influence of hydrophilic carbon dots on polyamide thin film nanocomposite reverse osmosis membranes, J. Membr. Sci. 537 (2017) 42–53.
[7] M.E. Ali, L. Wang, X. Wang, X. Feng, Thin film composite membranes embedded with graphene oxide for water desalination, Desalination 386 (2016) 67–76.
[8] N.K. Khanzada, et al., Removal of organic micropollutants using advanced membrane-based water and wastewater treatment: a review, J. Membr. Sci. 598 (2020) 117672.
[9] S.M. Abtahi, et al., Micropollutant rejection of annealed polyelectrolyte multilayer based nanofiltration membranes for treatment of conventionally-treated municipal wastewater, Sep. Purif. Technol. 209 (2019) 470–481.
[10] M.M. Damtie, et al., Removal of fluoride in membrane-based water and wastewater treatment technologies: performance review, J. Environ. Manag. 251 (2019) 109524.
[11] Y. Vasseghian, M. Alimohamadi, E.-N. Dragoi, C. Sonne, A global meta-analysis of phthalate esters in drinking water sources and associated health risks, Sci. Total Environ. (2023) 166846.
[12] H.N. Tran, N.B. Nguyen, N.H. Ly, S.-W. Joo, Y. Vasseghian, Core-shell Au@ ZIF-67-based pollutant monitoring of thiram and carbendazim pesticides, Environ. Pollut. 317 (2023) 120775.
[13] Y. Vasseghian, D. Sezgin, D.C. Nguyen, H.Y. Hoang, M.S. Yilmaz, A hybrid nanocomposite based on CuFe layered double hydroxide coated graphene oxide for photocatalytic degradation of trimethoprim, Chemosphere 322 (2023) 138243.
[14] K. Wang, J. Luo, Y. Wei, K. Wu, J. Li, Z. Chen, Practical application of machine learning on fast phase equilibrium calculations in compositional reservoir simulations, J. Comput. Phys. 401 (2020) 109013.
[15] G.N. Marichal Plasencia, J. Camacho-Espino, D. Ávila Prats, B. Peñate Suárez, Machine learning models applied to manage the operation of a simple swro desalination plant and its application in marine vessels, Water 13 (18) (2021) 2547.
[16] R. Rohani, M. Hyland, D. Patterson, A refined one-filtration method for aqueous based nanofiltration and ultrafiltration membrane molecular weight cut-off determination using polyethylene glycols, J. Membr. Sci. 382 (1–2) (2011) 278–290.
[17] Y. Yoon, P. Westerhoff, S.A. Snyder, E.C. Wert, J. Yoon, Removal of endocrine disrupting compounds and pharmaceuticals by nanofiltration and ultrafiltration membranes, Desalination 202 (1–3) (2007) 16–23.
[18] K. Boussu, et al., Characterization of polymeric nanofiltration membranes for systematic analysis of membrane performance, J. Membr. Sci. 278 (1–2) (2006) 418–427.
[19] R.R. Sharma, R. Agrawal, S. Chellam, Temperature effects on sieving characteristics of thin-film composite nanofiltration membranes: pore size distributions and transport parameters, J. Membr. Sci. 223 (1–2) (2003) 69–87.
[20] F.T. Liu, K.M. Ting, Z.-H. Zhou, "2008 eighth ieee international conference on data mining,", in: Eighth IEEE International Conference on Data Mining, 2008, pp. 413–422.
[21] A. Géron, Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, Inc., 2022.
[22] K. Naderi, M.S. Kalami Yazdi, H. Jafarabadi, F. Bahmanzadegan, A. Ghaemi, M. R. Mosavi, Modeling based on machine learning to investigate flue gas desulfurization performance by calcium silicate absorbent in a sand bed reactor, Sci. Rep. 14 (1) (2024) 954.
[23] L. Breiman, Random forests, Mach. Learn. 45 (2001) 5–32.
[24] H. Belyadi, A. Haghighat, Machine Learning Guide for Oil and Gas Using Python: A Step-by-step Breakdown with Data, Algorithms, Codes, and Applications, Gulf Professional Publishing, 2021.
[25] F. Bahmanzadegan, A. Ghaemi, Exploring the effect of zeolite's structural parameters on the CO2 capture efficiency using RSM and ANN methodologies, Case Studies in Chemical and Environmental Engineering 9 (2024) 100595.
[26] Z. Khoshraftar, A. Ghaemi, F.S. Taheri, Polyethylenimine-functionalized halloysite nanotube as an adsorbent for CO2 capture: RSM and ANN methodology, Current Research in Green and Sustainable Chemistry 7 (2023) 100389.
[27] A. Hemmati, A. Ghaemi, M. Asadollahzadeh, RSM and ANN modeling of hold up, slip, and characteristic velocities in standard systems using pulsed disc-and-doughnut contactor column, Separ. Sci. Technol. 56 (16) (2021) 2734–2749.
[28] L. Yang, A. Shami, On hyperparameter optimization of machine learning algorithms: theory and practice, Neurocomputing 415 (2020) 295–316.
[29] P. Mehrmohammadi, A. Ghaemi, Investigating the effect of textural properties on CO2 adsorption in porous carbons via deep neural networks using various training algorithms, Sci. Rep. 13 (1) (2023) 21264.
[30] F. Fathalian, S. Aarabi, A. Ghaemi, A. Hemmati, Intelligent prediction models based on machine learning for CO2 capture performance by graphene oxide-based adsorbents, Sci. Rep. 12 (1) (2022) 21507.
[31] M. Chahbani, D. Tondeur, Mass transfer kinetics in pressure swing adsorption, Sep. Purif. Technol. 20 (2–3) (2000) 185–196.